

Leo Rasmussen

AI Researcher and Intern

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About Me

I am a passionate researcher and electrical engineering and applied math student.

As a second-year student at CU Boulder, I am a senior in electrical engineering by hours set to finish my requirements by the end of this semester and set to finish a bachelor's and master's in applied math by the end of summer 2027.

My research focus is on extracting signal from complex data using statistical modeling of the dynamical systems underneath. This has many applications, and I am most interested in its use in quantitative finance as a way to more rigorously utilize and understand alternative data sources.

Education Progress and Plan

Jan 2027 – Aug 2027 **Master's of Science**- Applied Mathematics
College of Engineering and Applied Sciences - University of Colorado Boulder

Aug 2026 – Dec 2026 **Bachelor's of Science**- Applied Mathematics
College of Engineering and Applied Sciences - University of Colorado Boulder

Aug 2024 – May 2026 **Bachelor's of Science**- Electrical Engineering
College of Engineering and Applied Sciences - University of Colorado Boulder - GPA 3.9

Goal : PhD in applied mathematics

Project and Work Experience

Iterate.ai : Agentic AI Platform

May 2025 - Aug 2025

Intern

Key Areas: Agents, Model Context Protocol, RAG

- Software startup with years of serving customers
- Created dozens of MCP tools using several methods
- Worked on a mature codebase
- Used a combination of regex and AI tools to make entire product translatable into any language

Intrace.ai : AI Agents for Open Source Intelligence (OSINT)

Jan 2025 - Feb 2025

Intern

Key Areas: Agents, web scraping, intelligence

- Denver-based startup with VC funding provided by Verana Capital focusing on agentic AI
- Helped to build the backend to enable multi-branch investigations using specialized agents
- Most development in Python utilizing Agno (formerly Phidata) and several APIs

CU Boulder : Neural Networks and Autonomous Systems

July 2024 - Aug 2025

Supervision : Dr. Majid Zamani

Key Areas: Formal verification, reinforcement learning, systems

- Worked with Professor Zamani and PhD assistant to model and reproduce control barrier certificates using neural networks in Python
- Furthered research in the area of hybrid control systems and confidence guarantees
- Assisted with paper Transfer Learning for Control Systems via Neural Simulation Relations, submitted to NeurIPS
- Used specialized loss functions to provably guarantee model behavior

- Created a novel solution for aiding LLMs in causal reasoning
- Created a novel solution for automatic detection of personally identifiable information (PID)
- Developed and implemented model for detecting indirect PID utilizing Python, Pytorch, and vector embedding models
- Demonstrated proof of concept applicability

Notable Achievements

- SAT Score: 1590/1600, 800/800 on math
- (16) AP tests taken and passed prior to CU Boulder. (2) AP tests were self-study
- First place individually in Colorado Academic Decathlon competition 2023
- Almost 200 completed PicoCTF cybersecurity challenges
- 790 math subject GRE, 69th percentile, completely self-studied

Technical Skills

- **Highly proficient:** Python, C, Pytorch, Tinygrad, Transformers, Vector Embeddings, LSTMs, RAG, agents
- **Intermediate proficiency:** RISC-V assembly, Lean4, MCP, Jax, Pandas, Real Analysis, Abstract Algebra
- **Familiarity:** Haskell, ARM assembly, Java, CUDA, Quantum Mechanics, Neuroscience, Differential Geometry, Topological Data Analysis

Github References

Implicator

- Research project for implying information from given information
- Can be used in OSINT investigations to find out if information about an individual is probable
- Future versions can be used to see if certain events are probable given other information
- Originally used vector embeddings and neural networks, change to fine-tuning a Llama model
- Custom implementation of Llama-3.1-8B-Instruct (transformers library only used for tokenizer), custom implementation of LoRA, custom implementation of fine-tuning across two GPUs
- Intensive data processing, cleaning, and manipulation from an open source dataset (entirety of Wikipedia)

Ridge Functions Proof

- A formalization in Lean4 of the universal approximation theorem for ridge functions based on the "Ridge Functions" monograph by Alan Pinkus
- Crucial intermediate theorem for universal approximation theorem for neural networks (Leshno et al. 1993) and understanding neural networks formally
- Involved self-study of the field of ridge functions, approximation theory, use of modules and

polynomials to prove density

Mainline

- An autograd framework, written entirely in C
- Enough functionality to create a neural network
- Example code to train neural network on the MNIST dataset

KeyphraseScorer

- Uses multiple open source language models to score how well a keyphrase matches a given text
- Part of my work for Dr. Ravi Starzl before I became an intern
- Was put to use by the United States military
- Exact specifications can be found in the README file
- Production-level code, with documentation, error checking and logging, and example inputs and outputs

Compressor

- An LSTM model trained to arithmetically encode and decode text to take up less space
- Trained on the tiny_shakespeare dataset
- On average, the encoded text takes 30% fewer bits than the original
- Completely lossless text compression

ScratchCNN

- A convolutional neural network made using only numpy
- All derivatives calculated manually to understand the calculus of CNNs
- Used to train on MNIST